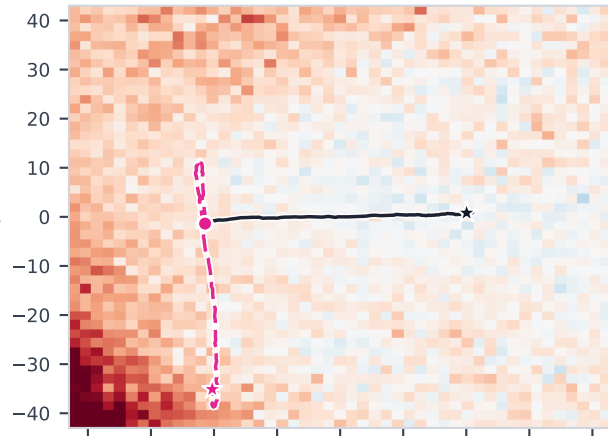


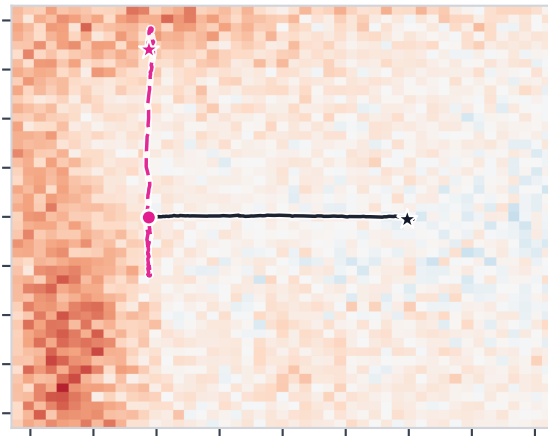
What learning adds: lift = $f(\theta)$ |RL on – $f(\theta)$ |RL off, same θ , same 5 maps — predator environment

Evolution — failed ($f < 0.136$, below viability)

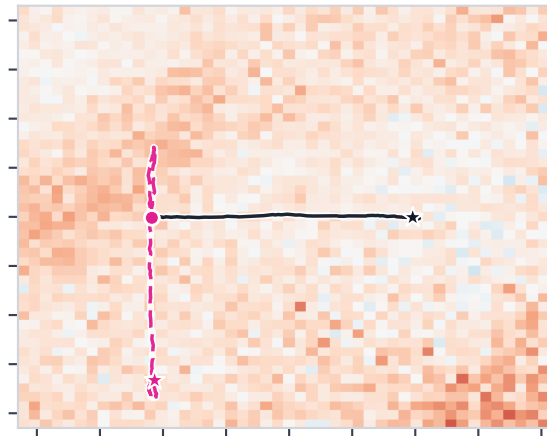
seed 2005 · run $f = 0.106$ · plane 69%
median lift +0.008



seed 2008 · run $f = 0.104$ · plane 69%
median lift +0.007



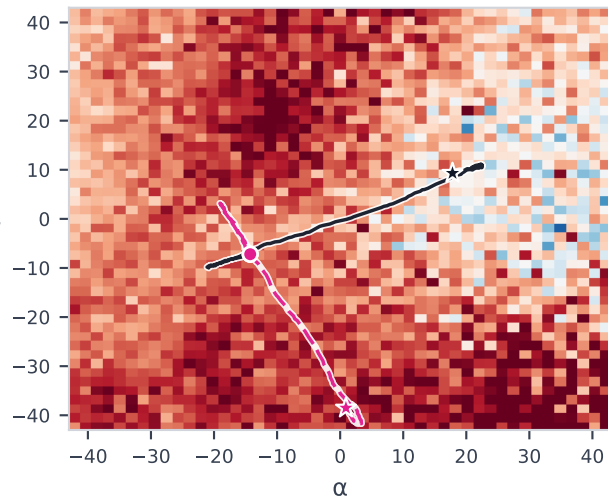
seed 2004 · run $f = 0.102$ · plane 69%
median lift +0.010



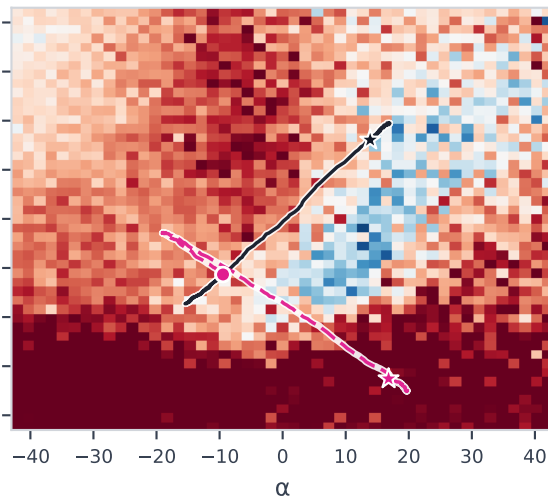
lift (food score added by in-lifetime learning) — normalised

Evolution — succeeded

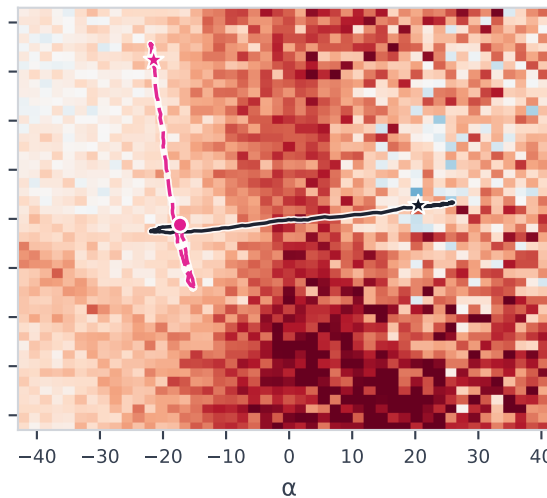
seed 2007 · run $f = 0.262$ · plane 62%
median lift +0.040



seed 2002 · run $f = 0.262$ · plane 62%
median lift +0.038



seed 2001 · run $f = 0.262$ · plane 64%
median lift +0.028



— evolution run - - - learning run ● gen 0 ★ final generation